Moving Seabins

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"Will adding efficient mobility to the existing static Seabins improve the surface litter collection by 63% in the next 2 years in the seas around Northern Ireland?"

Problem Scope

Northern Ireland's coastline has a growing problem with floating plastics in the seas on their coasts. This has a variety of negative impacts on environmental, economic and social growth in the area. The environmental impact of sea litter is the degradation of marine habitats, threatening marine life through entanglement, ingestion, and habitat destruction. Economically, the litter hampers coastal industries such as fishing and tourism, leading to financial losses and fewer opportunities for local workers. Socially, sea litter reduces the quality of life in coastal communities, hindering recreational activities and affecting the overall appearance of the coast.

Research

[Department of Agriculture, Environment and Rural Affairs 2022]

There is increasing surface litter in Northern Ireland and the static, land-based bin placement is inefficient. Current cleaning efforts currently require manual labour in combination with the Seabins. These levels of pollution and littering cause tourism profits to be affected.

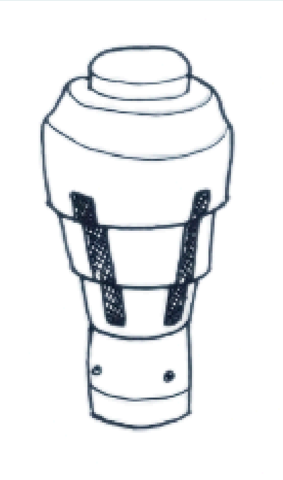
In This Document

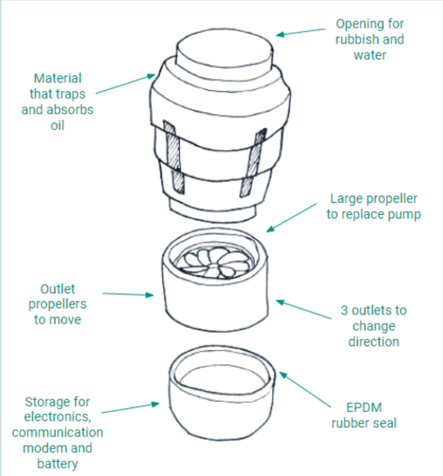
This following document will cover:

- The **general solution** to the problem
- The **overall design** of the product's hardware and software
- The **costs and feasibility** of this project
- The **project plan** and timeline

Hardware Design

General Design

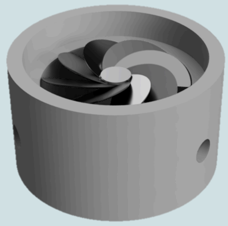




To the left are two images showcasing the product design:

- Sketch of product with all parts bolted together
- Exploded diagram of the Seabin with annotations

Here is also a 3D model of the propulsion system, used for material and manufacture considerations:



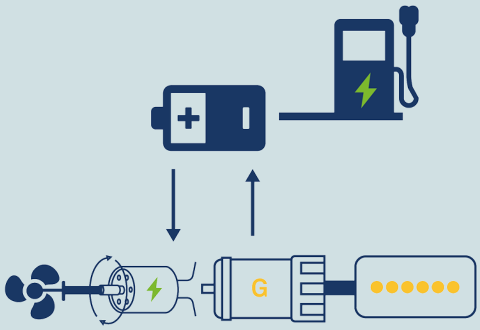
The rest of the section will break down each part of the design, and cover the considerations involved.

Propulsion System

The moving bins will be moved by **electric propellers** powered by electricity.

The electric propeller is composed of four parts:
motor, propeller, housing and sealing and control system

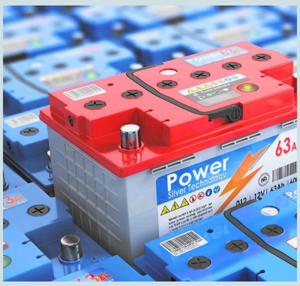
Considering the cost, the choice of motor is a **brushless DC motor**. This type of motor has a high initial cost but low maintenance costs, which makes the brushless DC motor **cheaper to operate in the long term**.



Battery Selection

We have chosen a **Lead Acid Battery** for our system.

A key aspect of the Seabin's functionality is its reliance on a reliable and efficient power supply. This electrical system powers various components critical to the Seabin's operation, from the water pump that draws in the contaminated water to the filtration mechanism that separates debris and pollutants. Understanding the intricacies of the power supply is fundamental to appreciating the Seabin's ability to contribute to marine conservation.



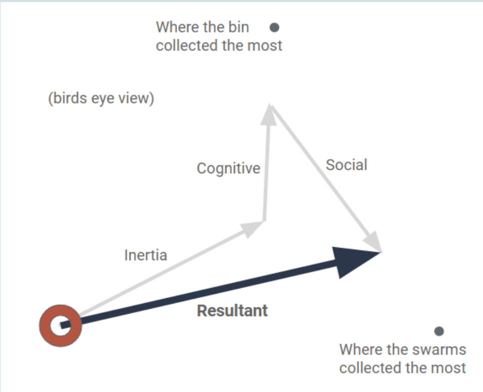
Battery properties:

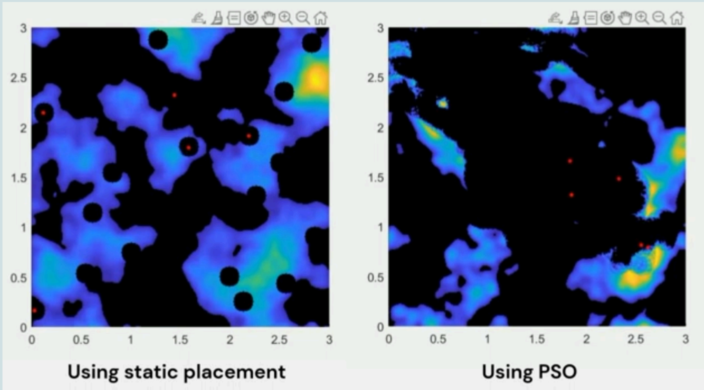
- A Seabin uses **110V / 220V (500 watts)** to operate
- The capacity of a lead-acid battery is **500Ah**
- Renewable energy storage, rechargeable, inexpensive, durable and reliable
- Life span is up to **8 years**
- Waterproofing (EPDM rubber)

Software Design

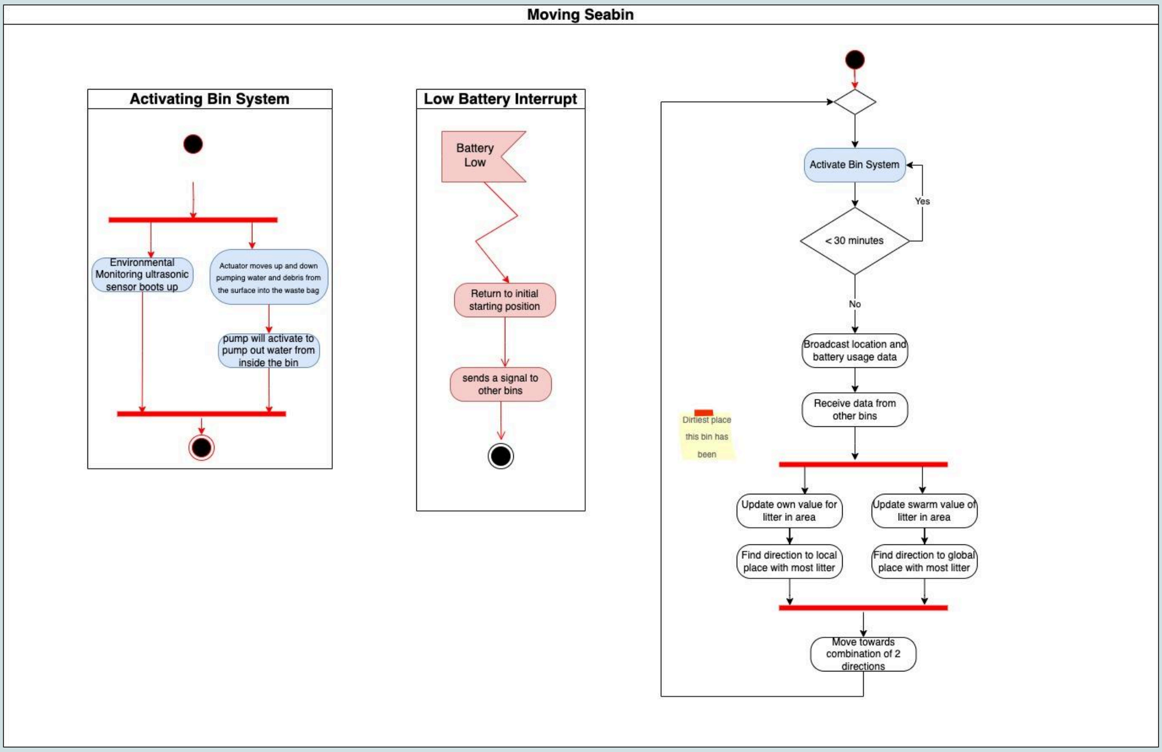
Algorithm Choice

Particle Swarm Optimization is an algorithm known for its simplicity and efficiency in finding solutions to optimization problems, but it's adaptable enough that we can tune it to work in this context. Another choice was ACO, which we will consider testing in the future.
A MATLAB simulation was built to test this, it was **>70% more efficient** than static placement





Activity Diagram

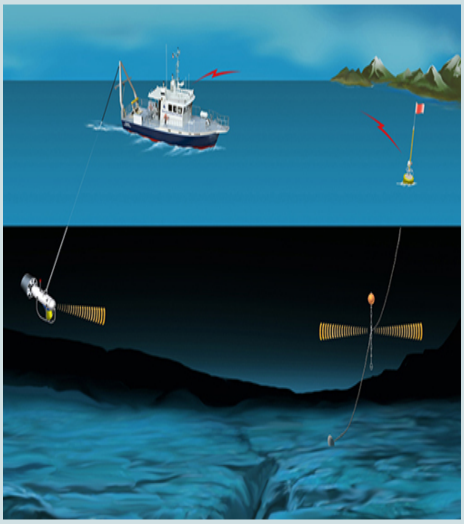


We will assume that a higher battery usage rate means the bin is heavier as it is using more power to remain stationary (it will need to use power for this as otherwise it will drift).
At any point, if the bin has a lower battery than the decided threshold (power needed to get back to dock plus a buffer), the low battery interrupt will be triggered and the bin will return to shore to be emptied and have its battery replaced by a newly charged one.

The Seabins will activate for 30 minutes at a time. All the Seabins will then broadcast their information on location and battery usage rate. We can use the change in rate of the battery usage in a given location to determine if there is a lot of waste in this area. This information will be used in the PSO algorithm mentioned before to calculate the direction the bin should move based on the local amount of waste and the global distribution of waste. It will find this new optimum location and start the waste collection process again. This process repeats..

Communication

Communication method



Acoustic communication was chosen, a few reasons why are:

- **Low power consumption**
  Acoustic communication reduces power usage, ensuring efficient operation of Seabins without draining battery life.
- **Effective over large distances**
  Acoustic signals can send signals effectively underwater, enabling seamless communication between Seabins even over long distances and around corners.
- **Well suited for underwater situations** Acoustic communication is specifically designed for underwater environments, making it a good choice for Seabins in aquatic settings and making it easy to implement.
- **Location and battery loss monitoring** Acoustic communication facilitates real-time monitoring of each Seabin's location and battery status, ensuring efficient maintenance and operation of the system.

Cybersecurity

Addressing Cybersecurity Issues in Acoustic Communication:

- **Preventing eavesdropping**
  Using strong encryption to keep messages private, and preventing others from listening in would help prevent eavesdropping.
- **Ensuring data Integrity**
  Use techniques like digital signatures and hash functions to make sure messages aren't changed by unauthorised users to keep data integrity.
- **Easy steps for better security**
  Keeping security measures up-to-date by regularly changing encryption codes and controlling who can access important information and communication channels within the Seabin network.

Cost Breakdown

Overview

Item	Initial Cost (£)	Power Consumption (W)	Cost (£/day)	Maintenance (£/year)
Seabin	£2,901.00	27	£0.19	£15.00
Communication modem	£50.00	0.204	negligible	£20.00
Battery	£314.00	n/a	n/a	£39.25
Material for casing	£10.00	n/a	n/a	£1.00
Casing construction	£10.00	n/a	n/a	n/a
Motors (1+3 small)	£50.00	440	£3.07	£11.00
Impeller (3)	£21.00	n/a	n/a	£20.00
Propeller	£25.00	n/a	n/a	£20.00
People to recharge/empty bins (One person per 50 bins)	n/a	n/a	£2.40	n/a
Total Added to bin	£480.00	440.204	£3.07	£111.25
Total for Seabin	£3,381.00	467.204	£3.26	£126.25

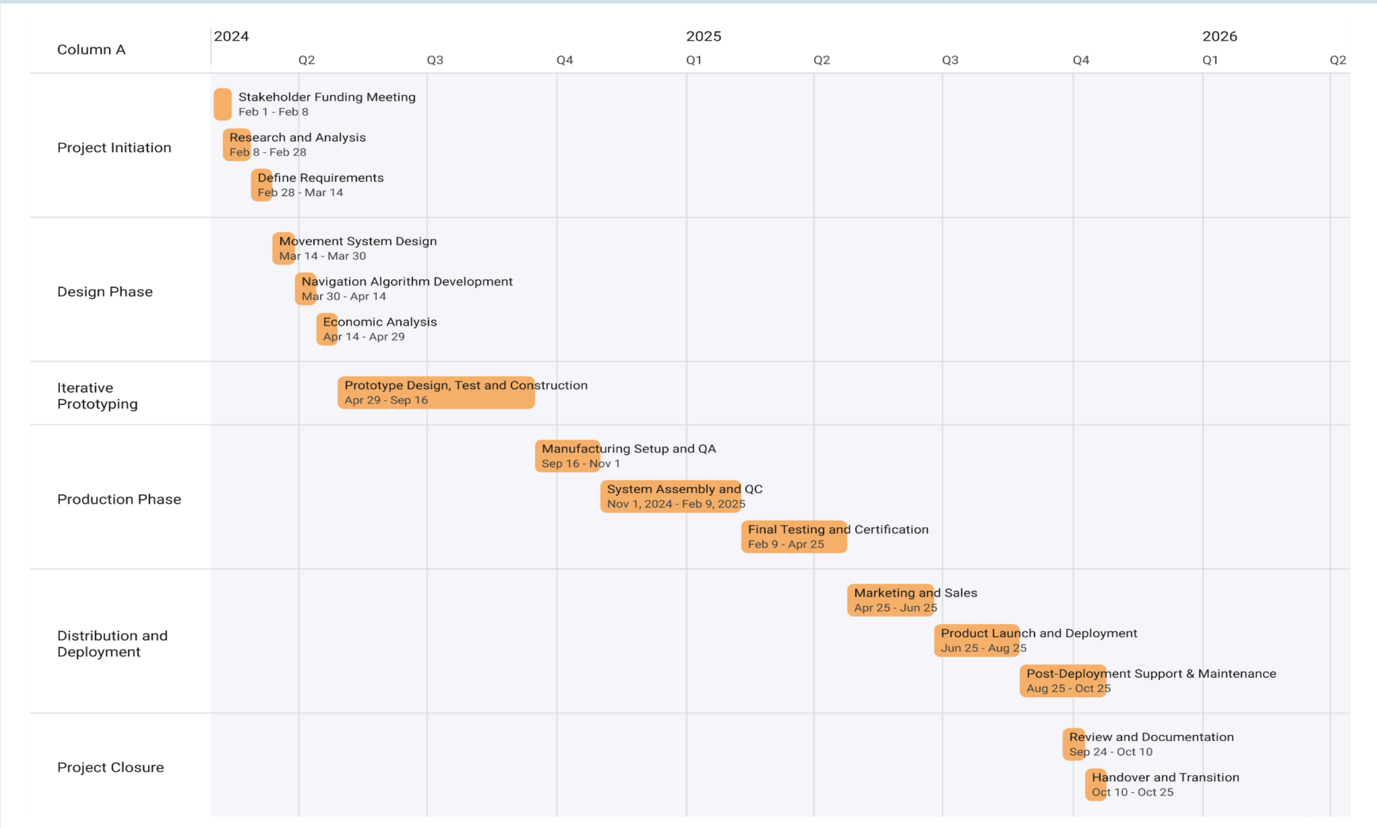
Funding and Stakeholders

The current Seabin Project has raised a lot of money through interest from large companies, government bodies and crowdfunding projects. Because of this, we believe these groups will also fund our improvement to the Seabins. Find out more at https://seabin.io/home/

Plan of Action

We have chosen an **iterative prototyping approach** as this is a low-risk way to manage the project. At each iteration, we will test the system and refine it according to our requirements.

Phase	Tasks
Project Initiation (Feb 2024 - Mar 2024)	- Stakeholder meeting and funding - Research of problem area and analysis of current system - Define objectives for the project
Design Phase (Apr 2024 - May 2024)	- Design movement algorithm (can be tested via simulations) - Design initial movement system - Initial prototype design
Prototype Development (Jun 2024 - Sep 2024)	- Construct prototype - Test and improve prototype - Iterate until idea and design are finalised
Production (Oct 2024 - May 2025)	- Manufacturing setup, organising suppliers and quality assurance - Assembly of system and quality control - Final testing and certification
Distribution & Deployment (Jun 2025 - Oct 2025)	- Marketing and sales to other potential customers - Product launch and deployment in Northern Ireland - Post-deployment support and ongoing maintenance
Project Closure (Nov 2025):	- Review and documentation (potential improvements) - Handover and transition - Celebration



Limitations

System	Physical
We have assumed that **battery usage** will be a good indicator of how much waste a bin has collected. However, this may not be a good measurement if a bin collects lots of **light, big items** as it will become full very quickly but indicate there is not a lot of waste in the area.	If the water is **too shallow**, the bottom of the bin may scrape the bottom. We have assumed that the bins can be collected in an area that is deep enough. **Sharp objects** may also cut the net and the bins can only collect **surface debris**.

Sustainability Commitment

Sustainability practices aim to use **resources efficiently, minimising waste** and **reducing the depletion of natural resources**. Sustainable practices such as reducing carbon emissions and promoting renewable energy, play a vital role in mitigating climate change and its associated impacts, such as rising temperatures, extreme weather events, and sea-level rise. We focus on prioritising sustainability in our project. Using **wind-generated energy** to generate power to recharge our batteries aligns with sustainability goals, reducing reliance on conventional energy sources. Waste collected that can be recycled **will be recycled**. The net that is being used to capture and store the waste collected is made from **recyclable plastic mesh**. Our idea is designed such that it is just an add-on to the current Seabin, this means that we can **reuse** the current Seabin and not manufacture a different structure of the Seabin. We aim to use materials that can be **responsibly disposed of or repurposed** at the end of their life cycle. The main structure of the bin will be made from polypropylene, since it can be re-melted and repurposed at the end of its life. Ideally, we could **use old casings to create new ones**. Polypropylene does not corrode easily and should have a long life, even in the sea. Our batteries would be lead-acid batteries since these can also be recycled. We will ensure a tight seal on the battery casings to ensure they do not leak any harmful chemicals.

Cybersecurity Statement

As well as the cybersecurity measures put in place for our communication systems, we will ensure that all data stored in the Seabins is kept secured. There will be firewalls in place to ensure location data and waste volume information is secured. The Seabins will not have any personal data and only stores information relating to locations, its battery charge and waste levels in the area.

Risks

Risk	Control measure
Issues with the electric propulsion system	We can update the algorithm to notify other Seabins if one Seabin is stuck and is not collecting any rubbish.
Eavesdropping on communications	We will encrypt any of the location information that is being transmitted.
Boats hitting the Seabins	We will make the Seabins a bright orange colour to warn boats of their presence. If they are hit, however, this should not be a problem as the bin is stable and buoyant.
Fish entanglement in Seabins	If the fish survive until the Seabins are back to shore they will get released

Overall risk level is low because we are using a pre-existing product. If the solution does not end up working, the loss will be low as the Seabins can still be used as static Seabins and the cost for the other materials is low.